

CSC227 TUTORIAL QUESTIONS

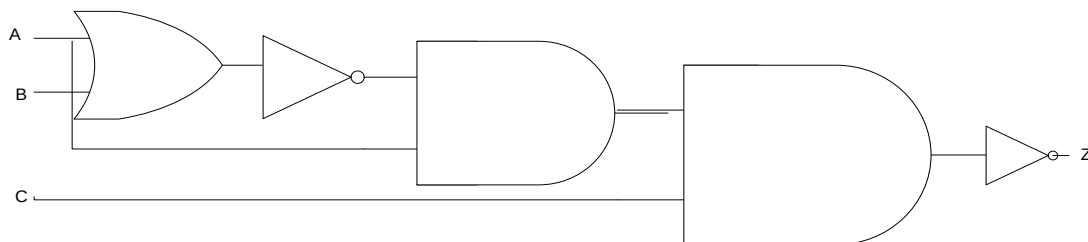
1. Minicomputers in the eighties had 18 address signals and 18-bit address bus. Answer the following questions:
 - i. What was the address space of these computers?
 - ii. What may have been the largest possible memory of these computers in bytes if the memory location is 1 Byte?
 - iii) How long should the program counter (PC) of these computers be?
2. i) What is pipelining? ii) Calculate how long it will take to execute on a regular **non-pipelined and pipelined** architecture. There are 4 operations with 5 stages each: IF, ID, EX, M and WB; stage1= 8ns, stage2 = 9ns, stage3 = 11ns, stage4 = 13ns, stage5 =16ns. Provide the Gantt Charts for both pipeline and non-pipeline operations. Also calculate the Speedup of the Pipeline system
3. i) Describe the three types of data hazards. ii) What type of data hazards do the following instructions belong?

MULTD F0, F2, F4
 DIVD F10, F0, F6
 2. DIVD F10, F0, F6
 ADDD F6, F8, F2
 3. MULTD F0, F2, F4
 SD F8, 50 (R5)

1.
4. Find the minimum sum-of-products represented by the K-map below:

x_1	00	01	11	10
x_3	1	0	1	1
	d	0	1	d

5. Using Boolean algebra, produce a simpler circuit from the circuit given below



6. i) What do you understand by addressing mode?

7. Use a **zero address, one address and two address instruction format** to write a Pentium 4 assembly language program segment for the expression below. For each format calculate the number of bytes required given that opcode is 2 bytes and memory access is one byte.

$$Z = V * ((U + W)/(X * Y));$$

8. Write down a signed decimal number with a value of -25 in 8-bit fixed point presentation in each of the three 8-bit modes for the representation of signed numbers.
9. i) Write the value of 4.75 in 32-bit floating point format **IEEE 754**.
 ii) The floating point value is written in the memory as 44FAC000(16). Which value does it represent if it is written in the format **IEEE 754**?
 ii) What do you understand by Instruction Format and Instruction Set?
10. i) State the difference between Von Neumann Architecture and Harvard Architecture (include diagrams)
 ii) What do you understand by Von Neumann Bottleneck? Suggest three ways to mitigate this bottleneck.
11. Complete the table below. Show all necessary work done to arrive at each answer. Use **10bits**.

Value	Unsigned	S-M	1's Complement	2's Complement

12. Use the reverse of the IEEE Single Precision to determine the value (Base 10) of the following hexadecimal (base 16) value: **BE7C2002H**
13. i) Describe three categories of Instructions that we have. Give two examples of each.
 ii) Discuss the characteristics of computer memory in terms of the following:
14. State Four (4) clear differences between RISC and CISC. Give two examples of processors in each category.
15. Determine the range of **POSITIVE and NEGATIVE** numbers that can be represented in a 16-bit register using **Sign-Magnitude, 1's complement and 2's complement**.
16. A processor can support a maximum memory of 4GB where the memory is word addressable (a word = 2 bytes). Determine the number of bits required to uniquely address the entire memory.

17. i) List four (3) differences between DRAM and SRAM?
 ii) Differentiate between Overflow and Underflow. Give one example for each.
18. Consider a 4-stage pipeline processor. The number of cycles needed by the four instructions T1, T2, T3, T4 in stages S1, S2, S3, S4 is shown below:

	S1	S2	S3	S4
T1	2	1	1	1
T2	2	3	2	2
T3	2	1	1	3
T4	1	2	2	2

- i) Draw the chart one iteration of the loop given below
 ii) Determine the number of cycles needed to execute the following loop 1mk
 for ($i = 1$ to 2) {T1; T2; T3; T4;}